

PAPER_1761_T_21CM_MINUS_289_MK

Daniel T. Murphy

PAPER_1761 — 21cm Dark Ages Temperature = $T_{21} = -D_{\text{phys}} \times A_5 \times \beta_i \times 2 = -289.39 \text{ mK (vs EDGES -289)}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Cosmology

Date: June 18, 2026

Status: CLOSED — 0.14% closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1333 (PAPER_132x-135x corpus) gives:

``

$T_{21} = -D_{\text{phys}} \times A_5 \times \beta_i \times 2 = -289.39 \text{ mK (vs EDGES -289)}$

``

Status: **0.14%**.

UQFF Closed Identity

``

$T_{21} = -D_{\text{phys}} \times A_5 \times \beta_i \times 2 = -289.39 \text{ mK (vs EDGES -289) } 0.14\%$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1333

- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)

- Calculator dispatch: `calculate_paradox({"paradox": "t_21cm_minus_289_mk"})`

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